

GENERATIVE AI

Basics to Intermediate Guide

For Fresher Learners | 2025-2026

What You Will Learn

AI Basics | Machine Learning | Deep Learning | GenAI Models | LLMs | Prompt Engineering | RAG | Tools & Roadmap | Q&A Practice

SECTION 1 — UNDERSTANDING AI FROM SCRATCH

1. What is Artificial Intelligence (AI)?

Think of AI as teaching a computer to think and make decisions, just like humans do.

Normally, a computer follows exact instructions (code). But with AI, it learns from examples and figures things out on its own.

Real-Life Analogy

Teaching a child vs. programming a robot. You don't give a child step-by-step instructions for everything — they watch, learn, and figure it out. AI works the same way.

Types of AI

Type	What It Does
Narrow AI	Does one specific task (like face recognition, Google Translate)
General AI	Can do anything a human can (still doesn't exist fully)
Super AI	Smarter than all humans combined (fictional for now)

Q&A — AI Basics

Q: What is AI in simple words?

A: AI is making computers smart enough to learn from data and make decisions without being told exactly what to do.

Q: Is ChatGPT an example of AI?

A: Yes! ChatGPT is an example of Narrow AI — specifically Generative AI that generates text based on what it learned.

Q: What is the difference between AI and a normal program?

A: A normal program follows fixed rules. An AI program learns rules from data on its own.

SECTION 2 — MACHINE LEARNING

2. What is Machine Learning (ML)?

Machine Learning is a type of AI where the computer learns from data instead of being programmed with rules.

Simple Analogy

Spam email filter. You show it 1000 spam emails and 1000 normal emails. It learns the pattern. Next time it sees a new email, it decides — spam or not — on its own.

3 Types of Machine Learning

Type	Meaning + Example
Supervised Learning	You give labeled data (input + correct answer). Example: Predict house price based on size.
Unsupervised Learning	You give data without labels. AI finds patterns itself. Example: Customer grouping.
Reinforcement Learning	AI learns by trial and error, getting rewards. Example: Game-playing AI like Chess.

Common ML Algorithms (Simple Definitions)

Algorithm	What it does
Linear Regression	Predicts a number. Example: Predict salary based on experience.
Logistic Regression	Predicts yes or no. Example: Will it rain tomorrow?
Decision Tree	Asks questions step by step to reach an answer. Like a flowchart.
Random Forest	Many decision trees voting together. More accurate.
K-Means Clustering	Groups similar things together without labels.
SVM (Support Vector Machine)	Draws a line to separate two groups of data.
K-Nearest Neighbors (KNN)	Finds the closest similar examples to make a prediction.

Q&A — Machine Learning

Q: What is the difference between AI and ML?

A: AI is the big concept of making machines smart. ML is a method inside AI where machines learn from data.

Q: What is 'training' in ML?

A: Training means showing the ML model a lot of data so it can learn patterns from it, like studying for an exam.

Q: What is overfitting?

A: When a model learns the training data too well (memorizes it) but fails on new data. Like a student who memorizes answers but can't handle new questions.

Q: What is underfitting?

A: When the model is too simple and doesn't learn enough from the data. Like a student who didn't study at all.

Q: What is a dataset?

A: A collection of data used to train or test an AI model. Example: 10,000 images of cats and dogs.

SECTION 3 — DEEP LEARNING & NEURAL NETWORKS

3. What is Deep Learning?

Deep Learning is a more advanced type of ML that uses Neural Networks — computer systems inspired by how the human brain works.

Analogy

Your brain has neurons connected to each other. When you see a dog, millions of neurons fire together to recognize it. Deep learning mimics this with artificial neurons in layers.

What is a Neural Network?

A Neural Network has 3 layers:

- Input Layer — takes in data (like a photo)
- Hidden Layers — processes and finds patterns (the 'thinking' layers)
- Output Layer — gives the final answer (cat or dog)

'Deep' Learning = many hidden layers = more complex thinking.

Common Deep Learning Models

Model	Used For
CNN (Convolutional Neural Network)	Image recognition — detects faces, objects in photos
RNN (Recurrent Neural Network)	Sequence data — text, speech, time-series
LSTM (Long Short-Term Memory)	Remembers long sequences — used in translation, speech
Transformer	Handles text with attention — backbone of ChatGPT, BERT
GAN (Generative Adversarial Network)	Generates fake but realistic images, videos

Q&A — Deep Learning

Q: What is the difference between ML and Deep Learning?

A: ML uses simple algorithms like decision trees. Deep Learning uses neural networks with many layers and needs much more data.

Q: What is a Transformer model?

A: A type of neural network that uses 'attention' to understand which words in a sentence are most important. Used in ChatGPT, BERT, and most modern AI.

Q: What is Attention Mechanism?

A: It helps the model focus on the most relevant words in a sentence. Like how you focus on keywords when reading quickly.

Q: What are parameters in a model?

A: Numbers inside the model that it adjusts during training to get better. More parameters = more learning capacity. GPT-4 has trillions of parameters.

SECTION 4 — GENERATIVE AI (THE MAIN TOPIC)

4. What is Generative AI?

Generative AI is a type of AI that creates new content — text, images, audio, video, or code — based on what it has learned from existing data.

Simple Analogy

Imagine a student who reads 1 million books. You then ask them to write a new story. They use everything they've learned to create something new. That's Generative AI.

What can Generative AI create?

Output Type	Example Tools
Text	ChatGPT, Claude, Gemini — write essays, emails, code
Images	DALL-E, Midjourney, Stable Diffusion — generate photos from text
Audio / Music	Suno, ElevenLabs — create songs or realistic voice
Video	Sora, Runway ML — generate videos from text description
Code	GitHub Copilot, Cursor — write and fix code automatically
3D Models	Point-E, Shap-E — generate 3D objects from text

How is Generative AI different from normal AI?

Normal AI (Discriminative)	Generative AI
Classifies or predicts existing data	Creates brand new data
Example: Is this email spam?	Example: Write me a new email
Outputs a label or number	Outputs text, image, audio, video
Example: Face recognition	Example: Generate a new face

Q&A — Generative AI

Q: What is Generative AI in one line?

A: AI that generates new content (text, image, video, code) by learning patterns from huge amounts of data.

Q: Is ChatGPT the same as Generative AI?

A: ChatGPT is a product built using Generative AI. GenAI is the technology; ChatGPT is one application of it.

Q: What makes Generative AI special in 2024-2025?

A: The scale — models trained on internet-scale data can now write better than humans, generate realistic images, and code full applications.

SECTION 5 — LARGE LANGUAGE MODELS (LLMs)

5. What is a Large Language Model (LLM)?

An LLM is a very large AI model trained on billions of words from the internet, books, and articles. It learns to predict and generate text.

Analogy

Imagine autocomplete on your phone — but trained on the entire internet and with millions of times more brainpower. That's an LLM.

How does an LLM work? (Super Simple)

- Step 1: Read billions of text documents during training
- Step 2: Learn patterns — what word comes after what word
- Step 3: During use, predict the next best word, token by token
- Step 4: Keep predicting until the full answer is formed

Famous LLMs

Model	Creator + Key Strength
GPT-4 / GPT-4o	OpenAI — most popular, great at reasoning and coding
Claude 3.5 Sonnet	Anthropic — safe, honest, great for long documents
Gemini 1.5 Pro	Google — very large context window, multimodal
LLaMA 3.3 70B	Meta — open-source, can run on your own machine
Mistral	Mistral AI — small but powerful open-source model
IBM Granite	IBM — enterprise-focused, used in WatsonX

Key LLM Concepts

Term	Simple Meaning
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Token	A small chunk of text. 'Hello' = 1 token. Most LLMs think in tokens, not words.
Context Window	How much text the model can 'remember' at once. Bigger = better.
Temperature	Controls creativity. Low = safe/predictable. High = creative/random.
Fine-tuning	Taking a pre-trained model and training it more on specific data.
Embeddings	Converting words into numbers (vectors) so AI can understand meaning.
Hallucination	When an AI confidently says something false. A major challenge in LLMs.

Q&A — LLMs

Q: What does 'Large' mean in LLM?

A: It refers to the number of parameters (billions to trillions) — the more, the smarter the model can be.

Q: What is a token?

A: A token is roughly 3/4 of a word. '1000 tokens' is about 750 words. LLMs process text in tokens, not characters.

Q: What is hallucination in AI?

A: When an AI makes up facts confidently. Example: AI says 'Einstein won the Oscar for Best Actor in 1997.' — completely false but said with confidence.

Q: What is temperature in LLMs?

A: A setting from 0 to 1 (or 2). 0 = always gives the same safe answer. 1 = more creative and varied. Use low for coding, high for creative writing.

Q: What is fine-tuning?

A: Taking a general LLM and training it further on specific data. Example: Fine-tuning GPT on medical records to make a medical AI assistant.

SECTION 6 — PROMPT ENGINEERING

6. What is Prompt Engineering?

Prompt Engineering is the skill of writing better instructions (prompts) to get better results from an AI model.

Analogy

Giving instructions to a genie. A vague wish = vague result. A precise, detailed wish = exactly what you want. Prompt engineering is the art of making the wish perfectly.

Types of Prompting

Technique	What it means + Example
Zero-Shot	Ask the AI to do something without any examples. 'Translate this to French.'
One-Shot	Give one example, then ask. 'Apple = fruit. Car = ?'
Few-Shot	Give 2-3 examples, then ask. More accurate than zero-shot.
Chain of Thought (CoT)	Ask AI to think step by step. 'Solve this, explain each step.'
Role Prompting	Tell AI to act as a role. 'You are an expert SQL developer. Write a query.'
System Prompt	A hidden instruction given before user messages to set AI behavior.

Good Prompt vs Bad Prompt

Bad Prompt	Good Prompt
Write something about AI	Write a 200-word beginner explanation of Generative AI for a college student with no tech background.
Fix my code	I am using Python. This function gives a KeyError on line 5. Here is the code: [paste]. Fix it and explain why.

Make me a resume

Create a resume for a fresher Computer Science student with Python, SQL, and an AI RAG chatbot project. Format: clean, ATS-friendly, 1 page.

Q&A — Prompt Engineering

Q: What is a system prompt?

A: A secret instruction given to the AI before the conversation starts, to set its role and behavior. Example: 'You are a helpful coding assistant. Always explain your code.'

Q: What is Chain of Thought prompting?

A: Asking the AI to explain its reasoning step by step. This makes the AI more accurate, especially for math and logic.

Q: Why is prompt engineering important?

A: Because the same AI model gives completely different quality answers depending on how you ask. A good prompt can 10x the quality of an answer.

SECTION 7 — RAG (RETRIEVAL AUGMENTED GENERATION)

7. What is RAG (Retrieval-Augmented Generation)?

RAG is a technique where an AI searches a knowledge base first, then uses that information to generate a more accurate answer.

Analogy

Imagine an AI that can open a book before answering your question. Instead of relying only on memory (training), it looks up the relevant pages first, then answers. That's RAG.

RAG Flow — Step by Step

- Step 1: User asks a question
- Step 2: System converts the question into a vector (embedding)
- Step 3: Searches a vector database for relevant documents
- Step 4: Top relevant documents are retrieved
- Step 5: LLM reads those documents + user question and generates an answer

RAG Components

Component	Role + Example Tool
Embedding Model	Converts text to numbers. Example: text-embedding-ada-002, LLaMA
Vector Database	Stores and searches embeddings. Example: ChromaDB, Pinecone, FAISS
LLM	Generates final answer. Example: LLaMA 3.3, GPT-4, Claude
Retriever	Finds relevant chunks from vector DB based on query similarity
Chunking	Breaking large documents into smaller pieces before storing

Why is RAG important?

- Reduces hallucination — AI uses real documents, not just memory
- Up-to-date answers — you can add new data without retraining
- Private/custom knowledge — feed your own docs, PDFs, databases

- Cost-effective — no need to fine-tune expensive models

Q&A — RAG

Q: What is the difference between RAG and fine-tuning?

A: Fine-tuning bakes knowledge into the model permanently (expensive). RAG retrieves knowledge at runtime from a database (flexible and cheaper).

Q: What is a vector database?

A: A database that stores text as numbers (vectors) and can find similar content using math. Example: ChromaDB, Pinecone, FAISS.

Q: What is embedding?

A: Converting text into a list of numbers (vector) that captures the meaning. Similar sentences have similar vectors.

Q: What is chunking in RAG?

A: Breaking large documents into smaller pieces (chunks) so the retriever can find the most relevant part, not the whole document.

SECTION 8 — TOOLS, FRAMEWORKS & ECOSYSTEM

8. GenAI Tools and Frameworks

Key Frameworks

Framework / Tool	What it does
LangChain	Build AI apps with LLMs — chains prompts, tools, memory together
LlamaIndex	Connect LLMs to your own data — great for RAG
Hugging Face	Largest hub for open-source AI models and datasets
Streamlit	Quickly build web apps for AI in Python — no frontend needed
Groq API	Super fast inference API for LLaMA and Mistral models
Ollama	Run open-source LLMs locally on your own computer
FastAPI	Build APIs in Python — often used to serve ML models

Cloud AI Platforms

Platform	Key Feature
OpenAI API	Access GPT-4, DALL-E, Whisper via API
Google Vertex AI	Deploy ML/GenAI models on Google Cloud
IBM WatsonX	Enterprise AI platform — great for IBM projects
AWS Bedrock	Access multiple LLMs on Amazon Cloud
Azure OpenAI	Microsoft version of OpenAI API — enterprise security

Python Libraries for GenAI

Library	Use Case
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openai	Connect to OpenAI/Groq APIs
transformers	Use Hugging Face models in Python
chromadb	Local vector database for RAG projects
sentence-transformers	Generate text embeddings easily
pandas	Data processing for ML/AI datasets
numpy	Math operations — backbone of all ML
scikit-learn	Classic ML algorithms in Python

Q&A — Tools

Q: What is LangChain?

A: A Python framework to build applications using LLMs. It helps connect prompts, memory, databases, and tools into one pipeline.

Q: What is Hugging Face?

A: The GitHub of AI — a platform where thousands of open-source AI models, datasets, and spaces are shared for free.

Q: What is Streamlit?

A: A Python library to turn your AI script into a web app in minutes. No HTML/CSS needed. Great for demos and prototypes.

SECTION 9 — GENERATIVE AI LEARNING ROADMAP

9. GenAI Learning Roadmap — Fresher to Intermediate

Phase	Topics to Learn	Resources / Practice
Phase 1 (Week 1-2) Beginner	Python basics, pandas, numpy AI vs ML vs DL concepts Types of ML (supervised/unsupervised) Neural Network basics	freeCodeCamp Python Kaggle Learn (free) Google ML Crash Course
Phase 2 (Week 3-4) Core GenAI	What is GenAI? LLMs (GPT, Claude, LLaMA) Tokens, embeddings, context window Using OpenAI / Groq API in Python	OpenAI Cookbook Groq API docs Hugging Face tutorials
Phase 3 (Week 5-6) Prompt Eng.	Zero-shot, Few-shot, CoT prompting System prompts and roles Prompt chaining Common interview questions on prompting	LearnPrompting.org Anthropic Prompt Library PromptingGuide.ai
Phase 4 (Week 7-8) RAG & Projects	Vector databases (ChromaDB, Pinecone) Embeddings with sentence-transformers Build a RAG chatbot Deploy with Streamlit	LangChain docs LlamaIndex tutorials Your own RAG project
Phase 5 (Week 9-10) Intermediate+	LangChain agents and tools Fine-tuning basics IBM WatsonX and Vertex AI overview MLOps and model deployment basics	IBM SkillsBuild Google Cloud Skills Boost FastAPI for model serving

SECTION 10 — TOP INTERVIEW Q&A (FRESHER LEVEL)

10. Most Important Q&A for Interviews

AI & ML Fundamentals

Q: What is the difference between AI, ML, and Deep Learning?

A: AI is the big idea of making smart machines. ML is one way to do it using data. Deep Learning is a more advanced ML method using neural networks.

Q: What is supervised vs unsupervised learning?

A: Supervised: You train with labeled data (input + correct answer). Unsupervised: No labels — AI finds patterns on its own.

Q: What is a training dataset and test dataset?

A: Training data is used to teach the model. Test data is used to check how well it learned — like practice questions vs real exam.

Q: What is the bias-variance tradeoff?

A: Bias: Model is too simple and misses patterns (underfitting). Variance: Model is too complex and memorizes noise (overfitting). You want a balance.

Generative AI & LLMs

Q: What is Generative AI? Give an example.

A: AI that creates new content. Example: ChatGPT generates text, DALL-E generates images, GitHub Copilot generates code.

Q: What is the Transformer architecture?

A: A neural network design based on attention mechanisms. It is the backbone of all modern LLMs including GPT, BERT, Claude, and LLaMA.

Q: What is prompt engineering?

A: The skill of writing effective instructions for AI models to get better and more accurate outputs.

Q: What is RAG and why is it used?

A: RAG = Retrieval Augmented Generation. Used to give LLMs access to external/custom knowledge without retraining. Reduces hallucination and keeps answers up-to-date.

Q: What is vector similarity search?

A: Searching for the most similar text by comparing their vector (number) representations. Used in RAG to find relevant documents.

Q: What is the difference between GPT-3.5 and GPT-4?

A: GPT-4 has more parameters, better reasoning, longer context window, and supports images (multimodal). GPT-3.5 is faster and cheaper.

Practical / Project-Based

Q: What technologies did you use in your RAG Chatbot project?

A: LLaMA 3.3 70B (LLM), Groq API (fast inference), ChromaDB (vector database), Streamlit (web UI), SQLite (conversation history).

Q: What is ChromaDB?

A: An open-source vector database used to store and retrieve text embeddings. Used in RAG pipelines to find relevant document chunks.

Q: What is an API and how does it relate to AI?

A: An API lets applications talk to each other. AI APIs (like OpenAI, Groq) allow you to use powerful AI models in your own apps without running them yourself.

Q: What is model inference?

A: Running a trained model to get predictions or outputs. When you type a message to ChatGPT and it responds, that is inference.

Trending Topics to Know

Q: What is multimodal AI?

A: AI that understands multiple types of data — text, image, audio, video together. Example: GPT-4o can see an image and describe it.

Q: What is an AI Agent?

A: An AI that can use tools, browse the web, run code, and complete multi-step tasks on its own without human help at each step.

Q: What is IBM WatsonX?

A: IBM's enterprise AI platform for building, deploying, and governing AI models. Used by companies for AI in business workflows.

Q: What is model quantization?

A: Reducing the size of an AI model by using lower precision numbers. Makes models faster and cheaper to run — example: running LLaMA on your laptop.

SECTION 11 — KEY TERMS GLOSSARY

11. Quick Glossary — Know These Terms

Term	Simple Meaning
AI (Artificial Intelligence)	Making computers smart enough to learn and make decisions
ML (Machine Learning)	AI learns from data instead of hard-coded rules
Deep Learning	ML using neural networks with many layers
LLM (Large Language Model)	AI model trained on huge text data to understand and generate text
GenAI (Generative AI)	AI that generates new content — text, images, video, code
Token	A small unit of text. ~1 token = 0.75 words
Embedding	Text converted to a list of numbers representing its meaning
Vector Database	Database that stores embeddings and finds similar ones by math
RAG	Retrieve + Generate — AI searches docs before answering
Prompt	The instruction or question you give to an AI model
Fine-tuning	Re-training a model on specific new data for a specific task
Hallucination	AI confidently stating incorrect facts
Context Window	Maximum amount of text an LLM can process at once
Temperature	Controls randomness of AI output. 0 = safe, 1 = creative
Inference	Running a trained AI model to get an output
Transformer	Neural network architecture using attention. Backbone of LLMs
Zero-shot	Asking AI to do a task with no examples provided

Few-shot	Giving 2-3 examples before asking AI to do a task
Agent	AI that can take actions and use tools autonomously
Multimodal AI	AI that handles text + image + audio together

FINAL NOTES & TIPS FOR FRESHERS

12. Final Tips for Your Journey

Tip 1 — Build Projects

Theory alone won't get you hired. Build at least 1-2 GenAI projects. A RAG chatbot, an image classifier, or an AI-powered web app speaks louder than certificates.

Tip 2 — Know Your Resume Projects

Every line in your resume is a potential interview question. Know your RAG chatbot inside out — tech stack, why you chose each tool, challenges faced.

Tip 3 — Use AI to Learn AI

Use ChatGPT, Claude, and Gemini every day. Ask them to explain concepts, quiz you, review your code. The best way to understand GenAI is to use it constantly.

Tip 4 — Focus on Concepts Over Syntax

Interviewers care more about whether you understand WHY something works than exact code syntax. Understand RAG, embeddings, and LLM behavior deeply.

Tip 5 — Stay Updated

GenAI moves fast. Follow Hugging Face blog, OpenAI blog, and Anthropic blog. Even 15 minutes/day of reading keeps you ahead.

All the best in your Generative AI journey!

You already have great projects — now deepen your understanding and keep building.